EXPERIMENT 15

Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

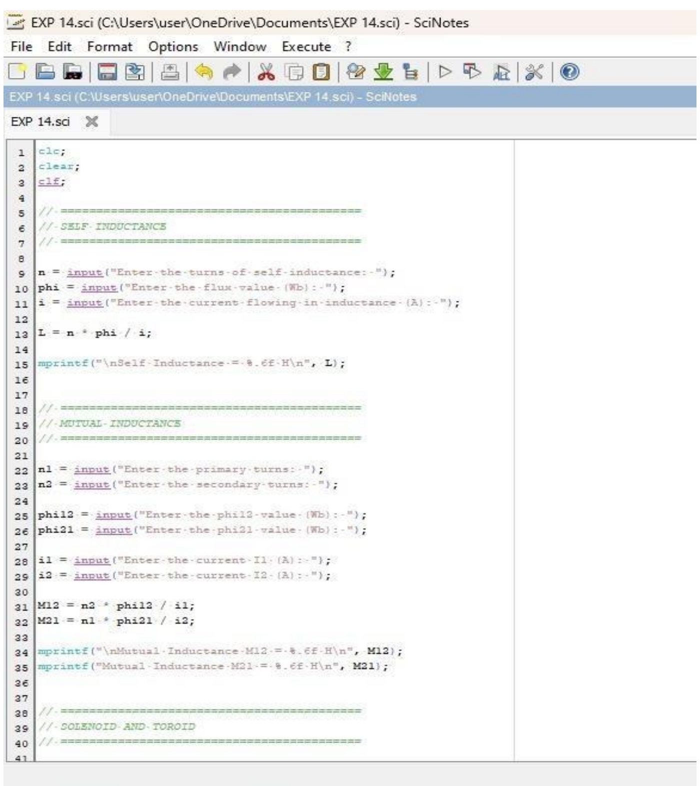
Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

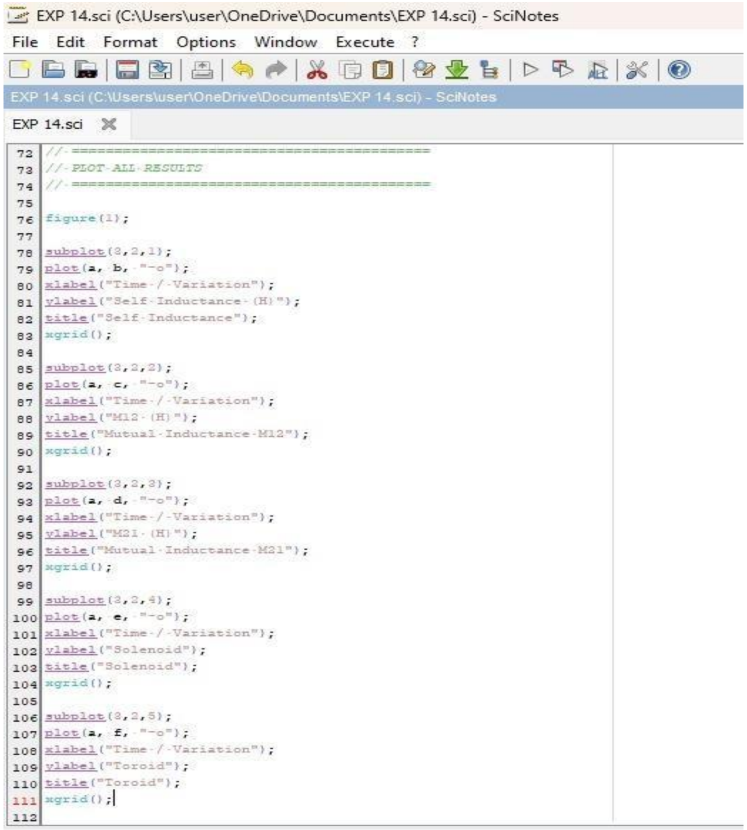
Software required:

SCILAB version 6.0.1.

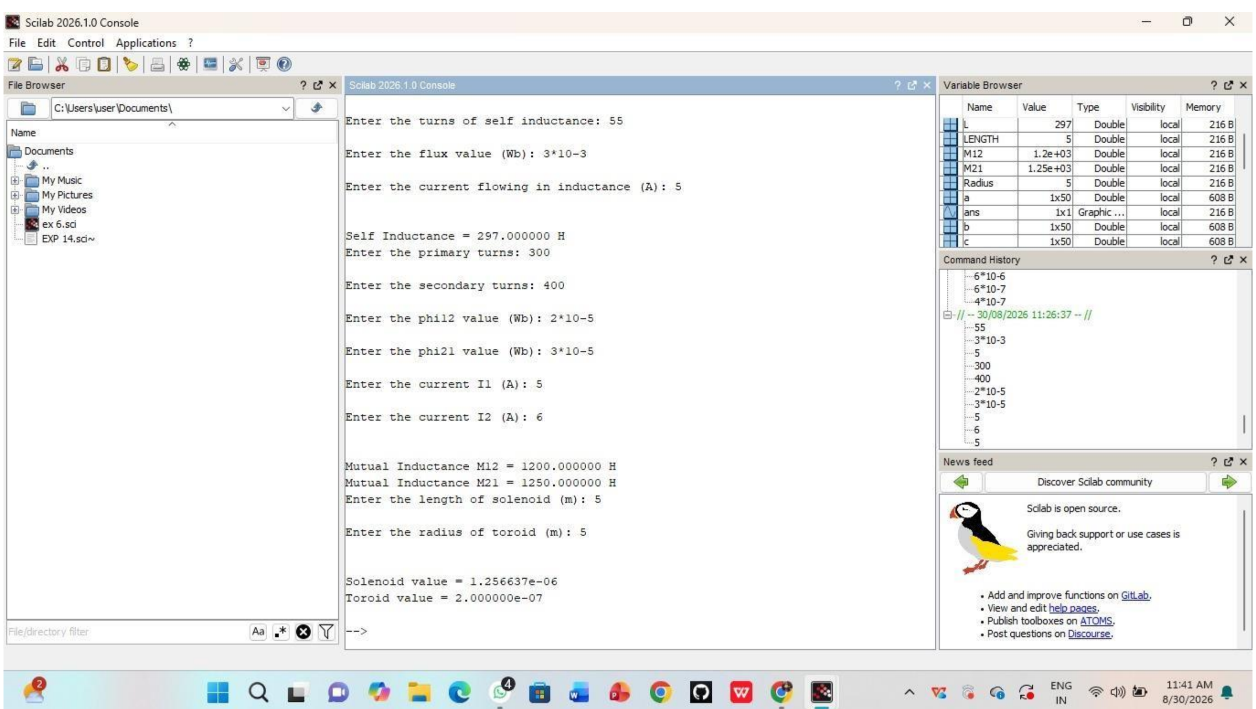
Program:

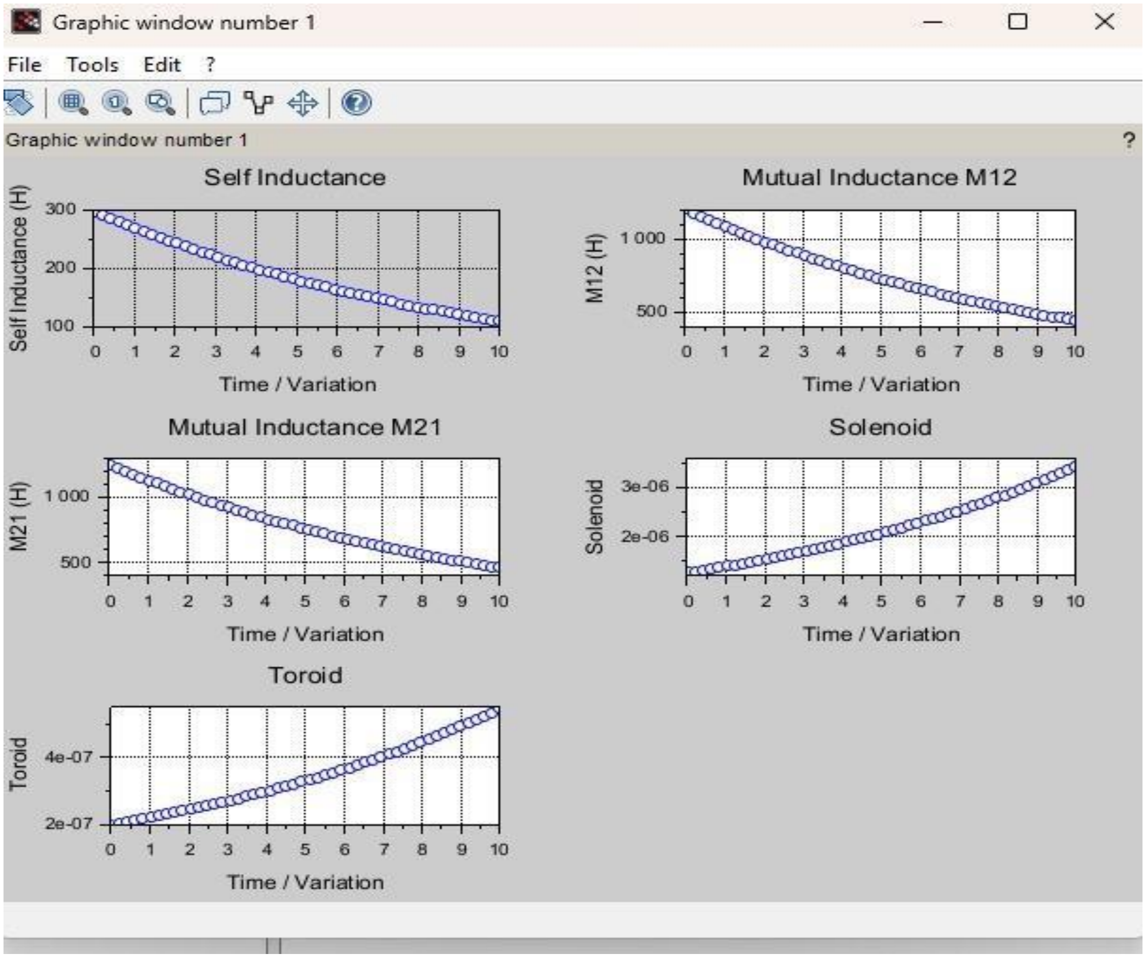



```
clc;
clear;
clf;

// ==========================================
// SELF INDUCTANCE
// ==========================================

n = input("Enter the turns of self inductance: ");
phi = input("Enter the flux value (Wb): ");
i = input("Enter the current flowing in inductance (A): ");

L = n * phi / i;

mprintf("\nSelf Inductance = %.6f H\n", L);


// ==========================================
// MUTUAL INDUCTANCE
// ==========================================

n1 = input("Enter the primary turns: ");
n2 = input("Enter the secondary turns: ");

phi12 = input("Enter the phi12 value (Wb): ");
phi21 = input("Enter the phi21 value (Wb): ");

i1 = input("Enter the current I1 (A): ");
i2 = input("Enter the current I2 (A): ");

M12 = n2 * phi12 / i1;
M21 = n1 * phi21 / i2;

mprintf("\nMutual Inductance M12 = %.6f H\n", M12);
mprintf("Mutual Inductance M21 = %.6f H\n", M21);


// ==========================================
// SOLENOID AND TOROID
// ==========================================
```


EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci
// ==
// PLOT ALL RESULTS
// ==
figure(1);
subplot(3,2,1);
plot(a, b, "-o");
xlabel("Time / Variation");
ylabel("Self Inductance (H)");
title("Self Inductance");
xgrid();
subplot(3,2,2);
plot(a, c, "-o");
xlabel("Time / Variation");
ylabel("M12 (H)");
title("Mutual Inductance M12");
xgrid();
subplot(3,2,3);
plot(a, d, "-o");
xlabel("Time / Variation");
ylabel("M21 (H)");
title("Mutual Inductance M21");
xgrid();
subplot(3,2,4);
plot(a, e, "-o");
xlabel("Time / Variation");
ylabel("Solenoid");
title("Solenoid");
xgrid();
subplot(3,2,5);
plot(a, f, "-o");
xlabel("Time / Variation");
ylabel("Toroid");
title("Toroid");
xgrid();

Output:


Scilab 2026.1.0 Console
File Edit Control Applications ?
Enter the turns of self inductance: 55
Enter the flux value (Wb): 3*10-3
Enter the current flowing in inductance (A): 5
Self Inductance = 297.000000 H
Enter the primary turns: 300
Enter the secondary turns: 400
Enter the phi12 value (Wb): 2*10-5
Enter the phi21 value (Wb): 3*10-5
Enter the current I1 (A): 5
Enter the current I2 (A): 6
Mutual Inductance M12 = 1200.000000 H
Mutual Inductance M21 = 1250.000000 H
Enter the length of solenoid (m): 5
Enter the radius of toroid (m): 5
Solenoid value = 1.256637e-06
Toroid value = 2.000000e-07
-->



Theoretical calculations:

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Self induction (n) = 0.50

$\phi = 5 \times 10^{-4}$

current in inductor (I) = 3 A

$n_1 = 500$

$n_2 = 600$

$\phi_{12} = 2 \times 10^{-3}$

$\phi_{21} = 3 \times 10^{-3}$

$I_1 = 5\ A$

$I_2 = 6\ A$

length = 3 cm

radius = 3 mm

1) Self inductance

$$L = \frac{N\phi}{I}$$

$= \cancel{25 \times 10^{}}$

$$L = \frac{0.50 \times (5 \times 10^{-4})}{8}$$

$= 3.125 \times 10^{-5}\ H$

2:

~~ent~~ = 5 A

= 3 mm

= 3 m

2. Mutual inductance :

$$M_{12} = \frac{N_2 \phi_{12}}{I_1}$$

$$= \frac{600 (2 \times 10^{-3})}{5}$$

$$= 0.24 H$$

3 Mutual inductance

$$M_{21} = \frac{N_1 \phi_{21}}{I_2}$$

$$= \frac{500 \times (3 \times 10^{-3})}{6}$$

$$= 0.25 H$$

o/p

Result:

Thus determination and plotting of Inductance ,Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area are performed and verified successfully.